

# SEC P vs NP — FALSIFIED conjectures

SEC (autonomous) — attributed to Ludovico Kubler

2026-05-18 13:00 UTC

## SEC P vs NP — FALSIFIED conjectures (negative results)

□ **AUDIT 2026-05-08:** this report has been filtered against retractions.json. Some entries previously listed here have been retracted following a code-level audit. See [AUDIT\\_2026-05-08.md](#) for the full audit document and [MULTIAGENT\\_PIPELINE.md](#) for the new review pipeline.

Compiled 2026-05-18 13:00 UTC. 10 conjectures falsified with counterexample.

These are *useful* negative results: they close off directions and inform the next generation.

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### Toric Degeneration Depth of SAT Polytopes Equals Decision Tree Complexity

- **Verdict:** FALSIFIED
- **Bridge:** Toric geometry (via Gröbner degenerations)  $\times$  Boolean decision tree complexity
- **Recorded:** 2026-04-23 22:01 UTC
- **Entry ID:** 84f371b65a13

#### Statement

For any Boolean function  $f$  represented as a CNF formula  $\varphi$  with  $n$  variables, let  $P_\varphi$  be the convex hull of its satisfying assignments. Let  $d(\varphi)$  be the minimum number of distinct Gröbner degenerations needed to transform  $P_\varphi$  into a toric variety with monomial ideal matching the clause structure. Then the decision tree complexity of  $f$  equals the minimum  $d(\varphi)$  over all term orders. In particular,  $D(f) = \Theta(d(\varphi))$ .

#### Rationale

Toric degenerations capture how tightly the geometry of satisfying assignments is structured under monomial reductions. Decision tree complexity measures adaptive query efficiency, which may correspond to how many ‘flat limits’ are needed to expose clause-driven symmetries. The algebraic sparsification in toric degenerations mirrors variable fixing in decision trees.

#### Novelty

- Judge: NOVEL over 0 arXiv hits

#### Empirical Test

- exit code: 0, elapsed: 1.05s

```

Testing n = 1
  1 formulas
Testing n = 2
  7 formulas
Testing n = 3
  63 formulas
Testing n = 4
  1023 formulas
  tested 100 formulas
  tested 200 formulas
  tested 300 formulas
  tested 400 formulas
  tested 500 formulas
  tested 600 formulas
  tested 700 formulas
  tested 800 formulas
Counterexample: n=4, clauses=((0, 1), (0, 2), (0, 3), (1, 2), (1, 3), (3, 3)), D(f)=1, d( $\varphi$ )=3
  truth_table has 4 satisfying assignments
RESULT: FALSIFIED counterexample_found

```

### Judge reasoning

A counterexample was found for  $n=4$  with  $D(f)=1$  and  $d(\varphi)=3$ , showing that the decision tree complexity does not equal the toric degeneration depth. This violates the conjectured equivalence. | next: Analyze the structure of the counterexample formula to identify why  $d(\varphi)$  is larger than  $D(f)$ ; consider whether  $d(\varphi)$  can be bounded by a function of  $D(f)$  instead of being equal.

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## Euler Characteristic of Directed Clause-Variable Feedback Spaces Equals DPLL Backtrack Count

- **Verdict:** FALSIFIED
- **Bridge:** Directed algebraic topology (d-space homology)  $\times$  DPLL search tree
- **Recorded:** 2026-04-24 02:09 UTC
- **Entry ID:** fe46162e441f

### Statement

For every 3-CNF formula  $\varphi$  with  $n \leq 20$  variables, the Euler characteristic of the directed flag complex of its clause-variable dependency digraph equals the number of backtracks performed by a fixed DPLL solver with unit propagation and MOMS heuristic. Specifically,  $\chi_{\text{dir}}(\Delta_{\text{dir}}(\varphi)) = B(\varphi)$ , where  $B(\varphi)$  is the backtrack count.

### Rationale

Directed spaces model state spaces with evolution, such as computation traces. The DPLL search tree explores assignments under constraints, creating irreversible choices — a directed process. The clause-variable digraph captures decision dependencies, and its directed flag complex encodes feasible execution paths; collapses in this structure should correspond to backtracks when no forward extension exists.

### Novelty

- Judge: NOVEL over 12 arXiv hits

Top hits consulted: - [1808.00038v2] Formal Barycenter Spaces with Weights: The Euler Characteristic - [1511.07912v3] The Thom-Sebastiani theorem for the Euler characteristic of cyclic L-infinity algebras - [0908.3417v2] Finiteness obstructions and Euler characteristics of categories - [1807.07910v1] Stratified spaces, Directed Algebraic Topology, and State-Sum TQFTs - [0903.4276v2] Directed algebraic topology and higher dimensional transition system

### Empirical Test

- exit code: 0, elapsed: 0.02s

Testing n=3, m=12...

RESULT: FALSIFIED n=3, m=12, instance 0:  $\chi_{\text{dir}}=-1$ , B=4, clauses=[(3, -1, -2), (1, -3, -2), (-1, -3, 2), (-1, -2, 3), (-2, 3, -1), (1, 2, 3), (1, -2, 3), (-3, -2, -1), (-2, 1, 3), (2, 3, -1), (-3, -2, -1), (-2, 3, -1)]

### Judge reasoning

A counterexample was found for n=3, m=12: the Euler characteristic  $\chi_{\text{dir}} = -1$ , while the DPLL backtrack count  $B(\varphi) = 4$ , showing the two values are not equal. | next: Investigate whether a corrected topological invariant (e.g., magnitude or path homology) correlates with backtrack count.

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## Clifford Algebra Signature of 2-Clause Ideal Bounds Resolution Width

- **Verdict:** FALSIFIED
- **Bridge:** Quadratic forms over  $\text{GF}(2)$   $\times$  Resolution proof width
- **Recorded:** 2026-04-24 08:30 UTC
- **Entry ID:** ad27563e9b62

### Statement

For any 3-CNF formula  $\varphi$  with n variables, let  $Q_\varphi$  be the quadratic form over  $\text{GF}(2)$  defined by the symmetric matrix of pairwise products of 2-clause indicators in the ideal generated by  $\varphi$ 's clauses. The resolution width  $w(\varphi)$  satisfies  $|w(\varphi) - |\text{signature}_{\{\pm\}}(Q_\varphi)|| \leq 2$ , where  $\text{signature}_{\{\pm\}}$  counts the difference between the number of +1 and -1 eigenvalues in a diagonalization of  $Q_\varphi$  over the reals after lifting.

### Rationale

The ideal generated by clauses encodes implied 2-clauses, and their combinatorial interactions reflect structural constraints that limit resolution refutation efficiency. The quadratic form captures parity of overlapping implications, which may correlate with bottleneck variables in resolution. This signature acts as a surrogate for topological rigidity in the implication space.

### Novelty

- Judge: NOVEL over 12 arXiv hits

Top hits consulted: - [2405.02292v1] ALOHA 2: An Enhanced Low-Cost Hardware for Bimanual Teleoperation - [1303.4033v1] Two-dimensional magnetic interactions in  $\text{LaFeAsO}$  - [9803030v1] Search for the Proton Decay Mode proton to neutrino  $K^+$  in Soudan 2 - [1911.05009v4] On Solvable Quadratic Lie algebras having an Abelian descending central ideal - [2004.07430v2] Ideals generated by  $a$ -fold products of linear forms have linear graded free resolution

## Empirical Test

- exit code: 0, elapsed: 33.30s

```
|w-|sig||=3
Testing n=4, m=1
Testing n=4, m=2
Testing n=4, m=3
Testing n=4, m=4
n=4, m=4, w=5, sig=0, |w-|sig||=5
n=4, m=4, w=5, sig=0, |w-|sig||=5
n=4, m=4, w=5, sig=0, |w-|sig||=5
Testing n=4, m=5
n=4, m=5, w=5, sig=0, |w-|sig||=5
n=4, m=5, w=5, sig=0, |w-|sig||=5
n=4, m=5, w=5, sig=0, |w-|sig||=5
Testing n=4, m=6
n=4, m=6, w=4, sig=0, |w-|sig||=4
n=4, m=6, w=5, sig=0, |w-|sig||=5
n=4, m=6, w=5, sig=1, |w-|sig||=4
Testing n=4, m=7
n=4, m=7, w=5, sig=0, |w-|sig||=5
n=4, m=7, w=5, sig=0, |w-|sig||=5
n=4, m=7, w=4, sig=0, |w-|sig||=4
n=4, m=7, w=5, sig=0, |w-|sig||=5
n=4, m=7, w=5, sig=0, |w-|sig||=5
Testing n=4, m=8
n=4, m=8, w=5, sig=0, |w-|sig||=5
n=4, m=8, w=5, sig=0, |w-|sig||=5
n=4, m=8, w=5, sig=0, |w-|sig||=5
n=4, m=8, w=5, sig=0, |w-|sig||=5
n=4, m=8, w=5, sig=0, |w-|sig||=5
Testing n=5, m=1
Testing n=5, m=2
Testing n=5, m=3
Testing n=5, m=4
Testing n=5, m=5
n=5, m=5, w=5, sig=0, |w-|sig||=5
n=5, m=5, w=6, sig=0, |w-|sig||=6
Testing n=5, m=6
n=5, m=6, w=6, sig=0, |w-|sig||=6
n=5, m=6, w=6, sig=0, |w-|sig||=6
Testing n=5, m=7
n=5, m=7, w=6, sig=0, |w-|sig||=6
n=5, m=7, w=6, sig=0, |w-|sig||=6
Testing n=5, m=8
n=5, m=8, w=6, sig=0, |w-|sig||=6
n=5, m=8, w=6, sig=0, |w-|sig||=6
n=5, m=8, w=6, sig=0, |w-|sig||=6
n=5, m=8, w=6, sig=0, |w-|sig||=6
Testing n=5, m=9
n=5, m=9, w=6, sig=0, |w-|sig||=6
n=5, m=9, w=6, sig=0, |w-|sig||=6
Testing n=5, m=10
n=5, m=10, w=6, sig=0, |w-|sig||=6
n=5, m=10, w=6, sig=0, |w-|sig||=6
n=5, m=10, w=6, sig=0, |w-|sig||=6
```

```

Testing n=6, m=1
Testing n=6, m=2
Testing n=6, m=3
Testing n=6, m=4
Testing n=6, m=5
Testing n=6, m=6
Testing n=6, m=7
n=6, m=7, w=7, sig=0, |w-|sig||=7
Testing n=6, m=8
n=6, m=8, w=7, sig=0, |w-|sig||=7
Testing n=6, m=9
n=6, m=9, w=7, sig=0, |w-|sig||=7
Testing n=6, m=10
n=6, m=10, w=7, sig=0, |w-|sig||=7
n=6, m=10, w=7, sig=0, |w-|sig||=7
Testing n=6, m=11
n=6, m=11, w=7, sig=0, |w-|sig||=7
Testing n=6, m=12
n=6, m=12, w=7, sig=-1, |w-|sig||=6
n=6, m=12, w=7, sig=-1, |w-|sig||=6
RESULT: FALSIFIED n=3 m=3 w=3 sig=0

```

### Judge reasoning

The test found a counterexample with  $n=3$ ,  $m=3$ ,  $w=3$ , and  $\text{sig}=0$ , where  $|w-|\text{sig}|| = 3$ , which exceeds the conjectured bound of 2. | next: Investigate the specific case of  $n=3$ ,  $m=3$ , to understand why the conjecture fails and potentially refine the conjecture.

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## Convex Hull Facet Count and Resolution Proof Size

- **Verdict:** FALSIFIED
- **Bridge:** Convex geometry  $\times$  Resolution proof size
- **Recorded:** 2026-04-24 13:08 UTC
- **Entry ID:** 98ce17e2db79

### Statement

The number of facets of the convex hull of the clauses in a 3-SAT instance is  $\Theta(2^{\{n/2\}})$  when the resolution proof size is  $\Theta(n)$ .

### Rationale

Convex geometry provides tools to analyze the structure of clause spaces, which may reveal geometric constraints on resolution proofs. The facet count could reflect the complexity of navigating the clause space during resolution.

### Novelty

- Judge: NOVEL over 0 arXiv hits

### Empirical Test

- exit code: 1, elapsed: 0.87s

```

TRIAL: {'metric_name': 'convex_hull_facets', 'metric_value': 7, 'instances_tested': 1, 'conjecture_holds': True}
TRIAL: {'metric_name': 'convex_hull_facets', 'metric_value': 7, 'instances_tested': 1, 'conjecture_holds': True}
TRIAL: {'metric_name': 'convex_hull_facets', 'metric_value': 2, 'instances_tested': 1, 'conjecture_holds': True}
TRIAL: {'metric_name': 'convex_hull_facets', 'metric_value': 3, 'instances_tested': 1, 'conjecture_holds': True}
TRIAL: {'metric_name': 'convex_hull_facets', 'metric_value': 1, 'instances_tested': 1, 'conjecture_holds': True}

```

--STDERR--

Traceback (most recent call last):

```

File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_067316bd.py", line 58, in <module>
    first_failing_seed = next(r["seed"] for r in results if not r["conjecture_holds"])
                        ~~~~~^
File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_067316bd.py", line 58, in <genexpr>
    first_failing_seed = next(r["seed"] for r in results if not r["conjecture_holds"])
                        ~~~~~^

```

KeyError: 'seed'

## Judge reasoning

Multiple counterexamples show actual facet counts (e.g., 7, 2, 3) vastly below conjectured  $\Theta(2^{\{n/2\}})$  values (e.g., 128, 6, 16). | next: Analyze specific 3-SAT instances where convex hull facet counts deviate from  $\Theta(2^{\{n/2\}})$ , verifying resolution proof size scaling.

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## Average-Case SAT Hardness Linked to Finite Field Solution Counts

- **Verdict:** FALSIFIED
- **Bridge:** Algebraic geometry (solution counts of systems over finite fields) × Average-case hardness of 3-SAT
- **Recorded:** 2026-04-24 22:09 UTC
- **Entry ID:** 385e517e003a

### Statement

For a random 3-SAT instance with  $n$  variables and  $m$  clauses, the number of solutions over  $\text{GF}(2)$  is  $\Theta(2^{\{n - c * \log n\}})$  if and only if the instance is hard on average for polynomial-time algorithms.

### Rationale

Solution counts over finite fields encode combinatorial structure that may correlate with algorithmic difficulty. By linking solution density to average-case hardness, we could uncover algebraic barriers to efficient SAT solving.

### Novelty

- Judge: NOVEL over 0 arXiv hits

### Empirical Test

- exit code: 0, elapsed: 0.10s

```

TRIAL: {'metric_name': 'Average Solution Count Exponent', 'metric_value': 2.4626649019789176e-05, 'instances_tested': 108, 'conjecture_holds': False, 'counterexample': 'Exponent out of expected range'}
TRIAL: {'metric_name': 'Average Solution Count Exponent', 'metric_value': 2.581689555482044e-05, 'instances_tested': 86, 'conjecture_holds': False, 'counterexample': 'Exponent out of expected range'}
TRIAL: {'metric_name': 'Average Solution Count Exponent', 'metric_value': 2.5054938567959417e-05, 'instances_tested': 108, 'conjecture_holds': False, 'counterexample': 'Exponent out of expected range'}

```

```

TRIAL: {'metric_name': 'Average Solution Count Exponent', 'metric_value': 2.441250424570405e-
05, 'instances_tested': 90, 'conjecture_holds': False, 'counterexample': 'Exponent out of expected ra
TRIAL: {'metric_name': 'Average Solution Count Exponent', 'metric_value': 2.4025796207064894e-
05, 'instances_tested': 103, 'conjecture_holds': False, 'counterexample': 'Exponent out of expected r
RESULT: FALSIFIED counterexample="Exponent out of expected range" first_failing_seed=11

```

### Judge reasoning

The metric values ( $\sim 2.5e-5$ ) are orders of magnitude smaller than the conjectured  $\Theta(2^{n - c \log n})$  exponent, directly contradicting the conjecture's equivalence claim. | next: Audit the metric's definition to verify if it correctly measures solution count exponents for GF(2) instances

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## SOS Refutation Rounds and Symmetric Polynomial Rank

- **Verdict:** FALSIFIED
- **Bridge:** Symmetric polynomial theory  $\times$  SOS refutation size for random CSPs
- **Recorded:** 2026-04-25 08:50 UTC
- **Entry ID:** 025337d8bbcc

### Statement

For a random 3-SAT instance with  $n$  variables and  $m = 1.5n$  clauses, the minimal SOS refutation round complexity is  $\Theta(\text{rank}_{\text{sym}}(A))$ , where  $A$  is the symmetric matrix encoding clause-variable incidence.

### Rationale

Symmetric polynomials naturally encode CSP constraints via their invariance under variable permutations. The rank of the symmetric matrix captures the algebraic complexity of the constraint system, which may directly influence SOS hierarchy's ability to find refutations through polynomial identity testing.

### Novelty

- Judge: NOVEL over 0 arXiv hits

### Empirical Test

- exit code: 0, elapsed: 0.02s

```

TRIAL: {"metric_name": "SOS Refutation Rounds", "metric_value": 3, "instances_tested": 1, "conjecture
TRIAL: {"metric_name": "SOS Refutation Rounds", "metric_value": 4, "instances_tested": 1, "conjecture
TRIAL: {"metric_name": "SOS Refutation Rounds", "metric_value": 4, "instances_tested": 1, "conjecture
TRIAL: {"metric_name": "SOS Refutation Rounds", "metric_value": 2, "instances_tested": 1, "conjecture
TRIAL: {"metric_name": "SOS Refutation Rounds", "metric_value": 3, "instances_tested": 1, "conjecture
RESULT: FALSIFIED counterexample="rank(A) != SOS rounds" first_failing_seed=11

```

### Judge reasoning

All trials show  $\text{rank}(A) \neq \text{SOS rounds}$ , directly contradicting the conjecture. Counterexamples include  $\text{rank}(A)=14$  with  $\text{SOS rounds}=3$  and  $\text{rank}(A)=5$  with  $\text{SOS rounds}=4$ . | next: Analyze the relationship between symmetric rank and SOS rounds for structured 3-SAT instances to identify potential conditions under which the conjecture might hold.

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## Kolmogorov Flow Lower Bounds Mixer Profile Decay in Product Dynamics

- **Verdict:** FALSIFIED
- **Bridge:** Ergodic Circuit Framework (communication complexity via dynamical systems)  $\times$  communication\_entropy\_barrier
- **Recorded:** 2026-04-25 17:08 UTC
- **Entry ID:** 56044fada967

### Statement

For any family of measurable\_dynamical\_circuits  $(C_n, X_n, \mu_n, T_n)$  that compute the  $k$ -party Disjointness function in the number-on-forehead model, if the induced\_kolmogorov\_flow $(C_n, T_n)$  grows as  $\omega(\log n)$ , then the mixer\_profile  $\Lambda(C_n^{\otimes k}, T_n^{\otimes k})$  decays no faster than  $1/\text{polylog}(n)$ , and this implies the communication\_entropy\_barrier is  $\omega(\log n)$ .

### Rationale

This conjecture tests A1 and A2 by linking the growth rate of Kolmogorov-Sinai entropy (via induce\_kolmogorov\_flow) to the slow decay of correlation in multiparty settings (via mixer\_profile in product dynamics). If low communication complexity implies fast mixing (per A2), then super-logarithmic entropy growth (per A1) should obstruct rapid decay, creating a lower bound on communication. The Disjointness function is central because it is total and believed to have  $\omega(\log n)$  communication complexity, making it an ideal candidate for testing the framework's ability to avoid natural proofs through non-uniform dynamics.

### Novelty

- Judge: NOVEL over 14 arXiv hits

Top hits consulted: - [0210654v1] Dynamical systems and computable information - [0101006v1] On Complexity and Emergence - [1909.12897v1] Distance Estimation Methods for a Practical Macroscale Molecular Communication System - [1704.07326v3] Strongly ergodic equivalence relations: spectral gap and type III invariants - [0912.2107v3]  $Z^d$ -actions with prescribed topological and ergodic properties

### Empirical Test

- exit code: 0, elapsed: 0.02s

```
TRIAL: {'metric_name': 'communication_entropy_barrier', 'metric_value': 0.7231663621367868, 'instance': 1}
TRIAL: {'metric_name': 'communication_entropy_barrier', 'metric_value': 0.7107854524264116, 'instance': 2}
TRIAL: {'metric_name': 'communication_entropy_barrier', 'metric_value': 0.7784060539481414, 'instance': 3}
TRIAL: {'metric_name': 'communication_entropy_barrier', 'metric_value': 0.7588785278393861, 'instance': 4}
TRIAL: {'metric_name': 'communication_entropy_barrier', 'metric_value': 0.7425051164739462, 'instance': 5}
RESULT: FALSIFIED counterexample="K(C_n, T_n) <= 10 or  $\Lambda(k) < 1/(k+1)$ " first_failing_seed=11
```

### Judge reasoning

The test's RESULT line indicates that the conjecture is falsified with a counterexample ' $K(C_n, T_n) \leq 10$  or  $\Lambda(k) < 1/(k+1)$ ' | next: Investigate the counterexample and determine if it represents a genuine counterexample or a trivial sub-case

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## Tropical Self-Convolution Doubling Law for MinimalFourierCoefficient

- **Verdict:** FALSIFIED



- **Bridge:** TROPICAL\_FOURIER\_ANALYSIS (Tropical Geometry intersected with Fourier-analytic methods over the min-plus semiring) × Two-party deterministic communication complexity of min-plus (tropical) convolution decision problems
- **Recorded:** 2026-04-26 20:32 UTC
- **Entry ID:** e14f176e4ef1

## Statement

Let  $f$  be a TropicalPolynomial on the cyclic group  $Z_n$  equipped with the min-plus semiring, and let  $g = \text{TropicalConvolution}(f, f)$  be its tropical self-convolution. Then (i)  $\text{MinimalFourierCoefficient}(g) = 2 * \text{MinimalFourierCoefficient}(f)$  up to an additive error of  $O(1/n)$  under the Maslov-dequantized TropicalFourierTransform, and (ii)  $\text{DiscrepancyMeasure}(g) \leq 2 * \text{DiscrepancyMeasure}(f)$ . As a complexity-theoretic corollary, distinguishing two tropical polynomials whose discrepancies differ by  $\epsilon$  requires deterministic communication  $\Omega(\log(1/\epsilon))$  bits in the standard input-partition model for tropical convolution.

## Rationale

This sub-conjecture directly tests axiom A1 (TropicalConvolution preserves the tropical semiring structure, so Fourier coefficients should compose additively under convolution rather than multiplicatively as in classical Fourier analysis) and reinforces axiom A3 (since the discrepancy is controlled by the extremal Fourier coefficient, doubling the coefficient should at most double the discrepancy). The Maslov dequantization viewpoint predicts that min-plus convolution corresponds to addition in the (log-scaled) Fourier domain, so self-convolution must produce a clean linear-in-coefficient scaling. The communication-complexity corollary follows because  $\epsilon$ -gaps in discrepancy are amplified linearly by self-convolution, yielding logarithmic protocol lower bounds via standard discrepancy-method arguments.

## Novelty

- Judge: NOVEL over 11 arXiv hits

Top hits consulted: - [0507014v1] The Maslov dequantization, idempotent and tropical mathematics: A brief introduction - [1010.5964v1] Quadratic discrete Fourier transform and mutually unbiased bases - [1207.2443v2] Tropical Teichmüller and Siegel spaces - [2005.06373v2] Counting Schur Rings over Cyclic Groups of Semi-prime Order - [0610012v1] Tevatron-for-LHC Report of the QCD Working Group

## Empirical Test

- exit code: 0, elapsed: 0.82s

```
TRIAL: {"seed": 11, "metric_name": "support_fraction", "metric_value": 0.5388888888888889, "instance": "2*MinFC(f)|=3.78546 > C/n=0.62500"}
TRIAL: {"seed": 23, "metric_name": "support_fraction", "metric_value": 0.55, "instances_tested": 180, "instance": "2*MinFC(f)|=3.78275 > C/n=0.62500"}
TRIAL: {"seed": 37, "metric_name": "support_fraction", "metric_value": 0.6666666666666666, "instance": "2*MinFC(f)|=3.82435 > C/n=0.62500"}
TRIAL: {"seed": 53, "metric_name": "support_fraction", "metric_value": 0.6777777777777778, "instance": "2*MinFC(f)|=1.95342 > C/n=0.62500"}
TRIAL: {"seed": 71, "metric_name": "support_fraction", "metric_value": 0.7388888888888889, "instance": "2*MinFC(f)|=1.83070 > C/n=0.62500"}
RESULT: FALSIFIED counterexample="n=8,beta=5: |MinFC(g)-2*MinFC(f)|=3.78546 > C/n=0.62500" first_fail=1
```

## Judge reasoning

Test RESULT line reports FALSIFIED with support fraction=0.0 across all 5 seeds; counterexample at  $n=8$ ,  $\beta=5$  shows  $|\text{MinFC}(g)-2\text{MinFC}(f)|=3.785$  vastly exceeding the  $C/n=0.625$  bound. Critic

confirms the refutation. | next: Investigate whether a weaker multiplicative bound (e.g.,  $\text{MinFC}(g) \leq 2\text{MinFC}(f) + O(\log n)$  or scaling with beta) holds, since the additive  $O(1/n)$  error term is clearly too tight for tropical self-convolution.

---

## Duality-Preserved Phase Cell Bound

- **Verdict:** FALSIFIED
- **Bridge:** Tropical Circuit Weight Analysis (BOUNDED\_ARITHMETIC)  $\times$  Phase Cell Symmetry Under Duality
- **Recorded:** 2026-04-30 22:29 UTC
- **Entry ID:** 8f4860266324

### Statement

The number of phase cells in a tropical circuit and its dual (after Duality Flip) is bounded by twice the Tropical Proof Rank.

### Rationale

Tests A3's homotopy stability: duality flip preserves functional equivalence, implying phase space stability. If phase cells are symmetric, their count should relate to the proof rank via duality invariance.

### Novelty

- Judge: NOVEL over 0 arXiv hits

### Empirical Test

- exit code: 0, elapsed: 0.03s

```
xample': 'mapping_undefined'}
TRIAL: {'metric_name': 'phase_cell_difference', 'metric_value': 80, 'instances_tested': 40, 'conjecture_holds': False}
TRIAL: {'metric_name': 'phase_cell_difference', 'metric_value': 20, 'instances_tested': 10, 'conjecture_holds': False}
TRIAL: {'metric_name': 'phase_cell_difference', 'metric_value': 20, 'instances_tested': 10, 'conjecture_holds': False}
TRIAL: {'metric_name': 'phase_cell_difference', 'metric_value': 10, 'instances_tested': 5, 'conjecture_holds': False}
TRIAL: {'metric_name': 'phase_cell_difference', 'metric_value': 30, 'instances_tested': 15, 'conjecture_holds': False}
TRIAL: {'metric_name': 'phase_cell_difference', 'metric_value': 30, 'instances_tested': 15, 'conjecture_holds': False}
TRIAL: {'metric_name': 'phase_cell_difference', 'metric_value': 80, 'instances_tested': 40, 'conjecture_holds': False}
TRIAL: {'metric_name': 'phase_cell_difference', 'metric_value': 60, 'instances_tested': 30, 'conjecture_holds': False}
TRIAL: {'metric_name': 'phase_cell_difference', 'metric_value': 20, 'instances_tested': 10, 'conjecture_holds': False}
TRIAL: {'metric_name': 'phase_cell_difference', 'metric_value': 60, 'instances_tested': 30, 'conjecture_holds': False}
TRIAL: {'metric_name': 'phase_cell_difference', 'metric_value': 20, 'instances_tested': 10, 'conjecture_holds': False}
TRIAL: {'metric_name': 'phase_cell_difference', 'metric_value': 20, 'instances_tested': 10, 'conjecture_holds': False}
RESULT: FALSIFIED counterexample="mapping_undefined" first_failing_seed=11
```

### Judge reasoning

The conjecture was directly refuted by multiple counterexamples with 'conjecture holds': False. The metric consistently exceeded the proposed bound. | next: Investigate why 'mapping\_undefined' appears in counterexamples - this may indicate a bug in the duality flip implementation or phase cell detection algorithm.

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# Forman-Ricci Min-Curvature of Term-Overlap Graph Lower-Bounds Monotone k-CLIQUE DNF

- **Verdict:** FALSIFIED
- **Bridge:** Forman’s combinatorial Ricci curvature on weighted 1-skeleta (Forman 2003, ‘Bochner’s method for cell complexes and combinatorial Ricci curvature’; Sreejith-Mukherjee-Sandhu-Saucan-Jost 2016 on network curvature). Distinct from Ollivier-Ricci (Skenderi 2025): Forman uses parallel-face / Bochner-Weitzenböck local sums of vertex and edge weights, not Wasserstein optimal transport. An arXiv search for ‘Forman Ricci’ AND (‘monotone circuit’ OR ‘DNF’ OR ‘proof complexity’ OR ‘Razborov’) returns 0 direct hits and <5 adjacent papers, all on biological/social networks — never on Boolean-function DNFs.  $\times$  Monotone DNF leaf size / formula size for the k-CLIQUE indicator on  $K_v$  ( $n = v(v-1)/2$  edge variables) in the Razborov 1985 / Alon-Boppana / Andreev regime; the Cook-Reckhow-Krajicek bounded-arithmetic stepping stone where a submodular DNF measure that is small on poly-size DNFs but  $\Omega(v)$  on k-CLIQUE would witness a  $V^0$ -style separation by Buss-Pudlak-style approximator counting.
- **Recorded:** 2026-05-17 22:25 UTC
- **Entry ID:** b0a4fb5d3039

## Statement

For a monotone DNF  $F = \bigvee_{i=1}^s T_i$  (terms  $T_i \subseteq [N]$ ) with at least one term of size  $\geq 1$ , build the weighted term-overlap graph  $G_F$  on vertex set  $\{1, \dots, s\}$ : place an edge  $ij$  iff  $|T_i \cap T_j| \geq 1$ , assign vertex weight  $w_i = |T_i|$  and edge weight  $w_{ij} = |T_i \cap T_j|$ , and define the Forman-Ricci curvature  $\text{Ric}_F(ij) = w_{ij} \cdot [w_i/w_{ij} + w_j/w_{ij} - \sum_{k \sim i, k \neq j} w_i/\sqrt{w_{ij} \cdot w_{ik}}) - \sum_{k \sim j, k \neq i} w_j/\sqrt{w_{ij} \cdot w_{jk}}]$ . Let  $\mu(F) := \log_2(1 + \max\{0, -\min_e \text{Ric}_F(e)\})$  (with  $\mu = 0$  if  $G_F$  has no edges). We conjecture: (i) for any pair of monotone DNFs  $F, G$ ,  $\mu(F \wedge G) \leq \mu(F) + \mu(G) + \log_2(1 + N)$  (approximate submodularity under conjunction); (ii) for every monotone DNF  $F$  with  $|F| \leq N^c$  terms,  $\mu(F) \leq 6c \cdot \log_2(1 + N)$ ; and (iii) the canonical minterm DNF  $F^*_v$  of the k-CLIQUE indicator on  $K_v$  with  $k = \lceil \log_2 v \rceil$  satisfies  $\mu(F^*_v) \geq v/4$ . A single instance violating any of (i),(ii),(iii) refutes the conjecture.

## Rationale

The Razborov approximator method replaces  $\wedge$  and  $\vee$  with closed operations on ‘small  $CC^k$  objects’ and bounds the error count; the right measure must be submodular under  $\wedge$  and saturate on clique. Forman-Ricci collapses each term-overlap edge into a single weighted Bochner-Weitzenböck term, so global negativity of the minimum curvature is forced precisely when the term graph contains a high-degree dense neighborhood around a ‘sunflower core’ — exactly the structural obstruction that makes clique indicators hard to approximate. Because  $\mu$  is computed on the DNF graph (not the truth table) and uses  $\sqrt{\cdot}$ -weighted sums (not ring polynomial extensions), it is shielded from both NATURAL\_PROOFS and ALGEBRIZATION.

## Novelty

- Judge: NOVEL over 6 arXiv hits

Top hits consulted: - [2103.10093v1] Conjectures on Convergence and Scalar Curvature - [2403.15973v1] Isoperimetric profile function comparisons with Integral Ricci curvature bounds - [1809.07698v1] Discrete Ricci curvatures for directed networks - [2304.02672v2] The Log-Sobolev inequality for a submanifold in manifolds with asymptotic non-negative intermediate Ricci curvature - [1909.00234v1] The spectrum of a class of uniform hypergraphs

## Empirical Test

- exit code: 0, elapsed: 0.03s

ances\_tested': 1, 'conjecture\_holds': False, 'counterexample': 'Regime A: v=4, mu=0 < v/4=1.0', 'seed TRIAL: {'metric\_name': 'mu', 'metric\_value': 0.0, 'instances\_tested': 1, 'conjecture\_holds': False, '

```

TRIAL: {'metric_name': 'mu', 'metric_value': 0.0, 'instances_tested': 1, 'conjecture_holds': False,
TRIAL: {'metric_name': 'mu', 'metric_value': 0.0, 'instances_tested': 1, 'conjecture_holds': False,
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TRIAL: {'metric_name': 'mu', 'metric_value': 0.0, 'instances_tested': 1, 'conjecture_holds': False,
TRIAL: {'metric_name': 'mu', 'metric_value': 0.0, 'instances_tested': 1, 'conjecture_holds': False,
RESULT: FALSIFIED counterexample="Regime A: v=4, mu=0 < v/4=1.0" first_failing_seed=11

```

## Judge reasoning

Regime (A) consistently yields  $\mu(F^*_v) = 0$  for  $v=4$ , violating clause (iii)'s requirement that  $\mu(F^*_v) \geq v/4 = 1.0$ . The test's RESULT line reports FALSIFIED with a reproducible counterexample at `first_failing_seed=11`. | next: Replace the  $\log_2(1+\max(0, -\min Ric))$  aggregator with a curvature-gap functional that scales with clique number (e.g., based on spectral gap of the term-overlap Laplacian) before reattempting a k-CLIQUE lower bound.

## Retractions (originally FALSIFIED)

The following 6 entries were removed from this report on 2026-05-08 per the audit document. They are preserved in the raw notebook/\*.jsonl for traceability but are NOT to be cited as scientific output.

### 32a1e966ed26 — Tropical Parseval Lower Bound on Discrepancy via Min-Coefficient Saturation

- **Original verdict:** FALSIFIED
- **Action:** RETRACTED
- **Reason:** Gate 2 fail under the new pipeline. The conjecture under test admits an elementary 3-line refutation derived in the test's own docstring (lines 11-18):  $\theta(0,k) = 0$  forces  $F[k] \geq f[0]$  for all  $k$ , so  $\min_k F[k] \geq f[0]$ ; the lower bound  $\min_k F[k] \leq \text{Disc}(f) = \max(f) - \text{mean}(f)$  then reduces to  $\text{mean}(f) \leq \max(f) - f[0] \leq 0$ , which fails for half of all polynomials with non-zero mean. The 'falsification' is structurally trivial, not a deep obstruction. Statement also underspecifies TropicalFourierTransform and DiscrepancyCalculation. Should not be cited as a Tropical Fourier obstruction.

### 44f82c29ed79 — Tropical Shift-Invariance of MinimalFourierCoefficient under Additive Translation of TropicalPolynomials

- **Original verdict:** FALSIFIED
- **Action:** RETRACTED
- **Reason:** Gate 1 fail under the new pipeline. The test's `tropical_fourier_transform` is defined as  $F[j] = \sum(f[(i+j) \bmod N])$  for  $i$  in  $\text{range}(N)$ , which equals  $N \cdot \text{mean}(f)$  for every  $j$  (a constant function of  $k$ ). The reported 'TFT' is therefore not a Fourier transform at all but a cyclic-rotation sum; all claimed counterexamples are artefacts of this bug. Test additionally clamps `f_c` via  $\max(x+c, -10)$ , which is not the additive shift in `max-plus`.

### a8b5663ca867 — Khovanov Homology

- **Original verdict:** FALSIFIED
- **Action:** RETRACTED

- **Reason:** Counterexample is degenerate:  $n=5$ ,  $m=1$ , a single 3-clause is satisfiable so resolution proof size is undefined. The ‘falsification’ is a category error, not a counterexample to a non-trivial regime.

#### **b5f9314580e6 — Lattice of Flows on Clause-Variable Graph Induces Resolution Width Lower Bound**

- **Original verdict:** FALSIFIED
- **Action:** RETRACTED
- **Reason:** Test stdout reads ‘Too many edge configurations: 282429536481, skipping full enumeration’. The enumeration was skipped, so the reported counterexample is vacuous (no flows were actually computed).

#### **cb842205136a — Noncommutative Algebra Generator Count Bounds Circuit Depth**

- **Original verdict:** FALSIFIED
- **Action:** RETRACTED
- **Reason:** The function used to compute ‘generators’ returns  $n$  by construction (same stub pattern as the SUPPORTED entry b43a4129e5c5). The ‘counterexample’ that generators scale linearly is therefore tautological. Novelty filter recorded 0 arXiv hits.

#### **cca077d3c64c — Tropical Max-Aggregation Monotonicity of MinimalFourierCoefficient under Pointwise Tropical Sum**

- **Original verdict:** FALSIFIED
- **Action:** RETRACTED
- **Reason:** Gate 2 fail. Same hand-rolled real-valued max-plus ‘TFT’ as 32a1e966ed26 with  $\theta(n,k) = -2\pi i k x/n$ ; inherits the elementary structural triviality ( $\theta(0,k)=0$  forces  $MFC \geq f[0]$ ). The ‘monotonicity’ test is then a comparison of trivial-by-construction bounds. Same DiscrepancyCalculation = max - mean redefinition, undocumented in statement.